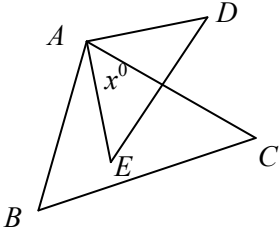


Question 1.**Marks**

- (a) Factorise fully: $a^2c - b^2c - abc^2 + ab$. 2
- (b) Evaluate: $\lim_{x \rightarrow 0} \frac{x \cos x}{\sin 3x}$. 2
- (c) Find the equation of the line that passes through the point of intersection of the lines $7x - 3y + 6 = 0$, $4x + 11y - 5 = 0$ and the point $(1, 1)$. 2
- (d) Given $G(x) = 3^{2x}$, find $G'(0)$. 2
- (e) The point P divides the interval AB externally in the ratio $1 : 2$. Find the coordinates of P when $A = (-2, 1)$ and $B = (1, -5)$. 2
- (f) Shade the region, on the Cartesian number plane, that satisfies both $y \leq \sqrt{9 - x^2}$ and $y \geq x^3$ simultaneously. 3
- (g) Solve for x : $3 \tan 2x = \sqrt{3}$, for $0 \leq x \leq \pi$. 2

Question 2. [START A NEW PAGE]

- (a) Simplify: $\sin(A + B) \sin(A - B) + \cos(A + B) \cos(A - B)$. 2
- (b) Given the function: $H(x) = x \left(\frac{2^x - 1}{2^x + 1} \right)$, show that $H(x)$ is an even function. 2
- (c) Solve for x : $|x - 3| < 1 + |x|$. 2
- (d)  In the diagram, triangles ABC and ADE are equilateral, and $\angle CAE = x^\circ$. 3
Not to scale
Copy the diagram onto your writing paper and Prove that: $\triangle BAE \equiv \triangle CAD$.
- (e) Find $\frac{dy}{dx}$ for: (i) $y = ex$. 1
(ii) $y = \sec 3x$. 1
- (f) Given $y = xe^{-x} + 3$, $y' = (1 - x)e^{-x}$ and $y'' = (x - 2)e^{-x}$,
(i) Show that the curve $y = xe^{-x} + 3$ has a point of inflection. 2
(ii) Find the equation of the inflectional tangent. 2

Question 3.	[START A NEW PAGE]	Marks
(a)	(i) Find the derivative of $\sqrt{1+6x}$.	1
	(ii) Hence, or otherwise, differentiate $x^3\sqrt{1+6x}$.	1
(b)	Find $\frac{dy}{dx}$ when: $y = \ln\left[\frac{x^4}{x-1}\right]$.	2
(c)	For what values of x is the curve: $y = \frac{5}{2}x^2 - \frac{1}{3}x^3 + 1$ increasing?	2
(d)	Solve for x : $\frac{4}{x^2-4} - \frac{1}{x-2} = 1$.	2
(e)	Show, by sketching, that: $\pi \sin x \geq 2x$ in the interval $0 \leq x \leq \frac{\pi}{2}$.	2
(f)	Solve for x , given that $a > 1$: $a^{2x} + a = (1+a)a^x$.	2
(g)	Find the simplified result of: $\frac{d}{dx}\left[\frac{x \sin x}{1 + \cos x}\right]$.	3

Question 4.

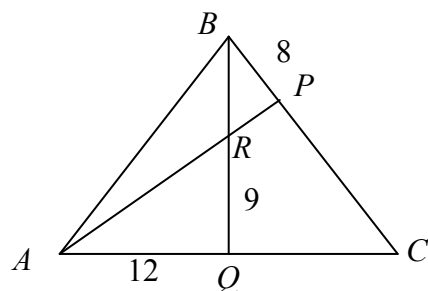
[START A NEW PAGE]

Marks

(a) Show that $\log_{a^n} 3 = \frac{1}{n} \log_a 3$, for $a > 0$ and $n \neq 0$.

1

(b) In $\triangle ABC$, $AP \perp BC$ at P and $BQ \perp AC$ at Q , as shown in the diagram.



Not to scale

Copy the diagram onto your writing paper and
(i) Prove that $\triangle AQR \parallel \triangle BPR$.

3

(ii) If $AQ = 12$, $QR = 9$ and $BP = 8$, find the area of $\triangle ABR$.

2

(c) Given the function: $f(x) = \frac{27(x-1)^2}{(x+1)^3}$.

(i) Show that: $f'(x) = \frac{27(x-1)(5-x)}{(x+1)^4}$.

2

(ii) Find the stationary points of the function $y = f(x)$.

2

(iii) Determine the nature of these stationary points.

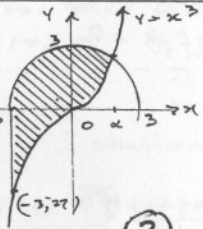
2

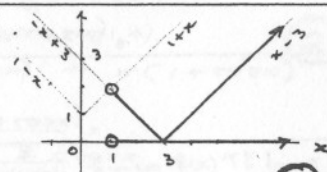
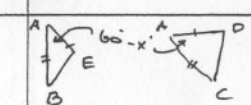
(iv) Sketch the curve $y = f(x)$, showing any asymptotes, intercepts and turning points.

3

THE END



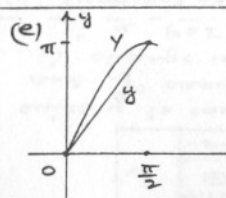
MATHEMATICS Extension 1 : Question 1		
Suggested Solutions	Marks	Marker's Comments
$1) a^2c - b^2c - abc^2 + ab$ $= a^2c - abc^2 + ab - b^2c$ $= ac(a-bc) + b(a-bc)$ $= (a-bc)(ac+b)$	1 1	
$2) \lim_{x \rightarrow 0} \frac{x \cos 3x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{3x}{3 \sin 3x} \cdot \cos x$ Method 1 $= \frac{1}{3} \cdot \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} \cdot \cos x$ $= \frac{1}{3} \times 1 \times \cos 0$ $= \frac{1}{3}$	1 1	other methods - Subst. $\theta = 3x$ - Small $\angle$ approx.
Let the line be: $7x - 3y + 6 + k(4x + 11y - 5) = 0$ Satisfies $\therefore 7 - 3 + 6 + k(4 + 11 - 5) = 0$ $10 + 10k = 0$ $\therefore k = -1$ so $7x - 3y + 6 - 1(4x + 11y - 5) = 0$ $\Rightarrow 3x - 14y + 11 = 0$	1 1	Point of intersection $\left(-\frac{51}{84}, \frac{59}{84}\right)$ $m = \frac{3}{14}$ $y - 1 = \frac{3}{14}(x - 1)$
$3) a(x) = 3^{2x} = 9^x$ $a'(x) = 3^{2x} \cdot \ln 9$ $a'(0) = 3^0 \cdot \ln 9 = \ln 9 = 2 \ln 3$	1 1	
A(-2,1) -1:2 B(1,-5) $P = \left(\frac{-1 \times 1 + 2 \times -2}{-1+2}, \frac{-1 \times (-5) + 2 \times 1}{-1+2} \right)$ $P = (-5, 7)$	1, 1	
 1/2 area curve region (c) $3 + \tan 2x = \sqrt{3}$ $\tan 2x = \frac{1}{\sqrt{3}}$ $0 \leq x \leq \pi$ $0 \leq 2x \leq 2\pi$ $\therefore 2x = \frac{\pi}{6} \text{ or } \pi + \frac{\pi}{6}$ $\therefore x = \frac{\pi}{12} \text{ or } \frac{7\pi}{12}$	1, 1	

MATHEMATICS Extension 1 : Question 2														
Suggested Solutions	Marks	Marker's Comments												
(a) $\sin(A+B) \sin(A-B) + \cos(A+B) \cos(A-B)$ $= \cos[(A+B) - (A-B)]$ $= \cos 2B$	1 1	Method 2: Expansion of LHS and $\sin^2 \theta + \cos^2 \theta = 1$												
(b) $H(-x) = (-x) \left(\frac{2^{-x} - 1}{2^{-x} + 1} \right) = -x \cdot \frac{(1 - 2^x)}{(1 + 2^x)}$ $= -x \cdot -1 \cdot \frac{(2^x - 1)}{2^x + 1} = x \cdot \frac{(2^x - 1)}{2^x + 1}$ i.e. $H(-x) = H(x)$ $\therefore H(x)$ is an EVEN FUNCTION	1 1													
(c) $x-3 < 1 + x$ Graphically: $-x + 3 = 1 + x$ $2 = 2x$ $\therefore x = 1$ $\therefore x > 1$ ✓ 	1 1	Algebraic: $ x-3 - x < 1$												
(d) In Δs BAE and CAD - AE = AD (given data) $\angle BAE = \angle CAD = (60 - x)^\circ$ (All angles in equilateral Δ are 60° each) - AB = AC (given) $\therefore \Delta BAE \cong \Delta CAD$ (SAS)	1	 For #1 or 3 For explaining #2 For SAS												
(e) (i) $y = e^x$ $\therefore \frac{dy}{dx} = e$	1	(ii) $y = \sec 3x$ $\therefore \frac{dy}{dx} = 3 \sec 3x \tan 3x$	1 each											
(f) (i) For possible points of inflection $y'' = 0$ $\therefore (x-2)e^{-x} = 0$ $\therefore x = 2 \text{ at } e^{-x} \neq 0$ $y = 2e^{-2} + 3$ TEST: <table border="1"> <thead> <tr> <th>x</th> <th>1.5</th> <th>2</th> <th>2.5</th> </tr> </thead> <tbody> <tr> <td>y''</td> <td>$\ominus$</td> <td>0</td> <td>$\oplus$</td> </tr> <tr> <td></td> <td>-0.11</td> <td></td> <td>0.04</td> </tr> </tbody> </table> Since y is cont + diffble over $1.5 \leq x \leq 2.5$ and y'' changes sign ($- \rightarrow +$) $\therefore$ change in concavity $\therefore$ a P.O.I. at $x = 2$ (ii) Gradient of Tangent: $m_T = -1e^{-2}$	x	1.5	2	2.5	y''	$\ominus$	0	$\oplus$		-0.11		0.04	1 1	$\therefore$ Equation $y - (2e^{-2} + 3) = -1e^{-2}(x - 2)$ $y = -e^{-2}x + 3 + 4e^{-2}$
x	1.5	2	2.5											
y''	$\ominus$	0	$\oplus$											
	-0.11		0.04											

MATHEMATICS Extension 1 : Question 3

Suggested Solutions	Marks	Marker's Comments
1) (i) $\frac{d}{dx} [\sqrt{1+6x}] = \frac{d}{dx} [(1+6x)^{\frac{1}{2}}]$ $= \frac{1}{2} (1+6x)^{-\frac{1}{2}} \times 6$ $= \frac{3}{\sqrt{1+6x}}$ ① Let $y = x^3 \sqrt{1+6x}$ $\frac{dy}{dx} = 3x^2 \sqrt{1+6x} + x^3 \times \frac{3}{\sqrt{1+6x}}$ $= \frac{3x^2(1+7x)}{\sqrt{1+6x}}$ ①	1	
2) $y = \ln \left[\frac{x^4}{x-1} \right]$ $= 4 \ln x - \ln(x-1)$ $\frac{dy}{dx} = \frac{4}{x} - \frac{1}{x-1} = \frac{3x-4}{x(x-1)}$ ②	1	
3) $y = \frac{5}{2}x^2 - \frac{1}{3}x^3 + 1$ $\frac{dy}{dx} = 5x - x^2$ for y to be increasing: $\frac{dy}{dx} > 0$ $\therefore 5x - x^2 > 0$ $x(5-x) > 0$ $\therefore 0 < x < 5$ to be increasing ②		1 For $5x - x^2 > 0$ 1 For $0 < x < 5$
4) $\frac{4}{x^2-4} - \frac{1}{x-2} = 1$ $e. \frac{4}{(x-2)(x+2)} - \frac{1}{x-2} = 1$ $\therefore x \neq \pm 2$ $\Rightarrow 4 - (x+2) = x^2 - 4$ $2 - x = x^2 - 4$ $\therefore x^2 + x - 6 = 0$ $(x+3)(x-2) = 0$ $\therefore x = -3 \text{ or } 2$ but $x \neq 2$ $\therefore x = -3$ only ②	$\frac{1}{2}$ 1 $\frac{1}{2}$	Note: $\frac{2-x}{x^2-4} = 1$ $\therefore \frac{-1}{x+2} = 1, x \neq 2$ processing.

MATHEMATICS Extension 1 : Question 3 CONTINUED

Suggested Solutions	Marks	Marker's Comments
(e)  $y = \pi \sin x$ $y = 2x$ $0 \leq x \leq \frac{\pi}{2}$ $\therefore y \leq y$ ②	1	sketching each correctly.
(f) $a^{2x} + a = (1+a)a^x$ $a^{2x} - (1+a)a^x + a = 0$ $(a^x - 1)(a^x - a) = 0$ $a^x = 1 \text{ or } a^x = a$ $\therefore x = 0 \text{ or } x = 1$ ②		Let $u = a^x$
(g) $\frac{d}{dx} \left[\frac{x \sin x}{1 + \cos x} \right]$ $= \frac{(1 \cdot \sin x + x \cos x)(1 + \cos x) - x \sin x (-\sin x)}{(1 + \cos x)^2}$ i.e. $= \frac{(\sin x + x \cos x)(1 + \cos x) + x \sin^2 x}{(1 + \cos x)^2}$ $= \frac{\sin x + \sin x \cos x + x \cos x + x \cos^2 x + x \sin^2 x}{(1 + \cos x)^2}$ $= \frac{\sin x(1 + \cos x) + x \cos x + x \cdot 1}{(1 + \cos x)^2}$ $= \frac{\sin x(1 + \cos x) + x(1 + \cos x)}{(1 + \cos x)^2}$ $= \frac{x + \sin x}{1 + \cos x}$ ③	2	

MATHEMATICS Extension 1 : Question 4

Suggested Solutions

Marks

Marker's Comments

$$\log_a 3 = \frac{\log 3}{\log a} = \frac{\log 3}{n \log a} = \frac{\log 3}{n}$$

①

(i) In Δs AQR and BPR

1. $\angle AQR = \angle BPR = 90^\circ$ (data)

2. $\angle ARQ = \angle BRP$ (vertically opposite angles are equal)

3.

$\therefore \Delta AQR \parallel \Delta BPR$ (equiangular)

②

$\frac{AR}{PR} = \frac{BR}{PR}$ (corresponding sides in similar triangles are in proportion)

$\therefore PR = 6$

$\therefore AR = 15$ (Pyth. Triad $\langle 3, 4, 5 \rangle$)

$BR = 10$ (" " " $\langle 3, 4, 5 \rangle$)

$$\text{Area ABR} = \frac{1}{2} \times 12 \times (9 + 10) - \frac{1}{2} \times 12 \times 9 = 60 \text{ square units}$$

③

$$f(x) = \frac{27(x-1)^2}{(x+1)^3}$$

$$f'(x) = \frac{27[2(x-1)(x+1)^3 - (x-1)^2 \times 3(x+1)^2]}{(x+1)^6}$$

$$= \frac{27(x-1)(x+1)^2[2(x+1) - 3(x-1)]}{(x+1)^6}$$

$$= \frac{27(x-1)[2x+2-3x+3]}{(x+1)^4}$$

$$= \frac{27(x-1)(5-x)}{(x+1)^4} \text{ qed.}$$

②

For S.P.s to occur $f'(x) = 0$

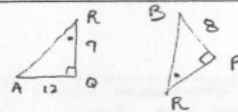
$$\therefore (x-1)(5-x) = 0$$

$$\therefore x = 1 \text{ or } 5$$

$$\text{or } y = 0 \text{ or } 2$$

$\therefore$ Stationary points are $(1, 0)$ and $(5, 2)$

②



MATHEMATICS Extension 1 : Question 4 cont.

Suggested Solutions

Marks

Marker's Comments

(iii) TEST NATURE.

at $x = 1$

x	0	1	2
$f'(x)$	$\frac{27(-)(+)}{(+)}$ -135	0	$\frac{27(+)(+)}{(+)}$ 1

Since $f(x)$ is cont + diffble over $0 \leq x \leq 2$ and $f'(x)$ changes sign $(- \rightarrow +)$

$\therefore$ a RELATIVE MIN T.P. at $(1, 0)$

②

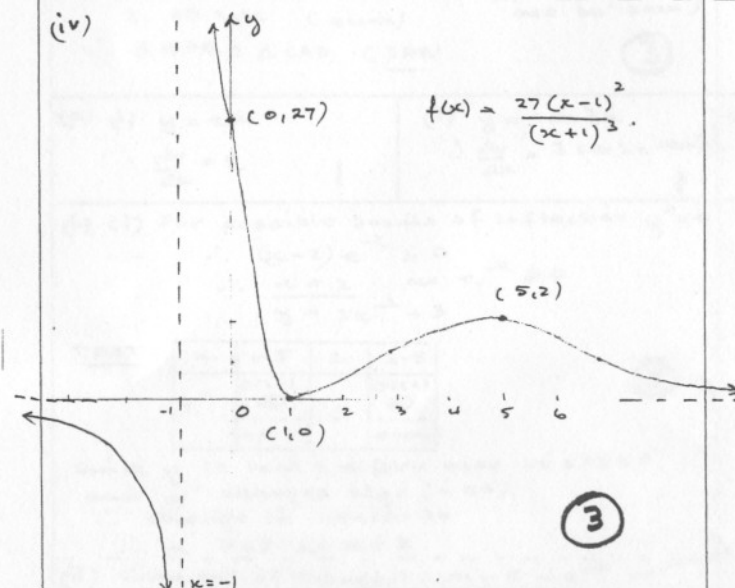
at $x = 5$

x	4	5	6
$f'(x)$	$\frac{27(+)(+)}{(+)}$ 0.12...	0	$\frac{27(+)(-)}{(+)}$ 0.056...

Since $f(x)$ is cont + diffble over $4 \leq x \leq 6$ and $f'(x)$ changes sign $(+ \rightarrow -)$

$\therefore$ RELATIVE MAX.T.P. at $(5, 2)$

(iv)



1 For indicating

VA $x = -1$
HA $y = 0$

1 For Min $(1, 0)$
Max $(5, 2)$

1 For correct shape and $(0, 27)$

* CHECK C.F.R. // C.F.E.

③